Advanced Programming Practice · 21CSC203P

Unit 5 · Session 6 · SLO 2 — Solved Worksheet

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Unit â€“ 5 : Session â€“ 6 : SLO â€“ 2 | Symbolic Limits and Series

■ I. Questions & Answers

1. lim (xâ†'âˆž) 1/x = 0 â€” as x grows without bound, 1/x shrinks to 0. In SymPy: limit(1/x, x, oo) â†' 0.

2. Sum of 1 + 1/2 + 1/4 + â€¦ â€” a geometric series with first term a = 1 and ratio r = 1/2; sum = a/(1 âˆ' r) = 1/(1 âˆ' 1/2) = 2.

3. lim (xâ†'0) sin(x)/x = 1 â€” the classic standard limit. In SymPy: limit(sin(x)/x, x, 0) â†' 1.

■ II. Code

• from sympy import symbols, limit, sin, oo, summation
• from sympy.abc import x, n
• print(limit(1/x, x, oo)) # 0
• print(summation((1/2)**n, (n, 0, oo))) # 2
• print(limit(sin(x)/x, x, 0)) # 1